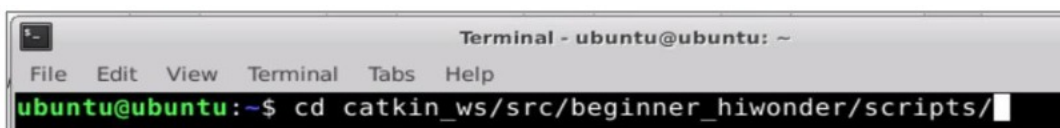


第 9 课 编写简单的服务端 Server

1.创建服务器代码

注意：在创建服务器代码之前，需要先创建工作空间和功能包，具体操作过程可前往目录“第2章 ROS开发入门->ROS基础课程->第3课 创建工作空间与功能包”查看文档。

- 1) 输入指令“`cd catkin_ws/src/beginner_hiwonder/scripts/`”，回车。



```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ cd catkin_ws/src/beginner_hiwonder/scripts/
```

- 2) 输入指令“`vi turtle_command_server.py`”编辑程序，复制下面程序。如需修改，再按下“`i`”即可修改。修改完成，按下“`Esc`”，输入“`: wq`”保存并退出。



```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder/scripts
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$ vi turtle_command_server.py
```

```
#!/usr/bin/env python

# -*- coding: utf-8 -*-

import rospy

import thread,time

from geometry_msgs.msg import Twist

from std_srvs.srv import Trigger, TriggerResponse

pubCommand = False;

turtle_vel_pub = rospy.Publisher('/turtle1/cmd_vel', Twist, queue_size=10)

def command_thread():

    while True:
```

```

if pubCommand:

    vel_msg = Twist()

    vel_msg.linear.x = 0.5

    vel_msg.angular.z = 0.2

    turtle_vel_pub.publish(vel_msg)

    time.sleep(0.1)

def commandCallback(req):

    global pubCommand

    pubCommand = bool(1-pubCommand)

    rospy.loginfo("Publish turtle velocity command![%d]", pubCommand)

    return TriggerResponse(1, "Change turtle command state!")

def turtle_command_server():

    rospy.init_node('turtle_command_server')

    s = rospy.Service('/turtle_command', Trigger, commandCallback)

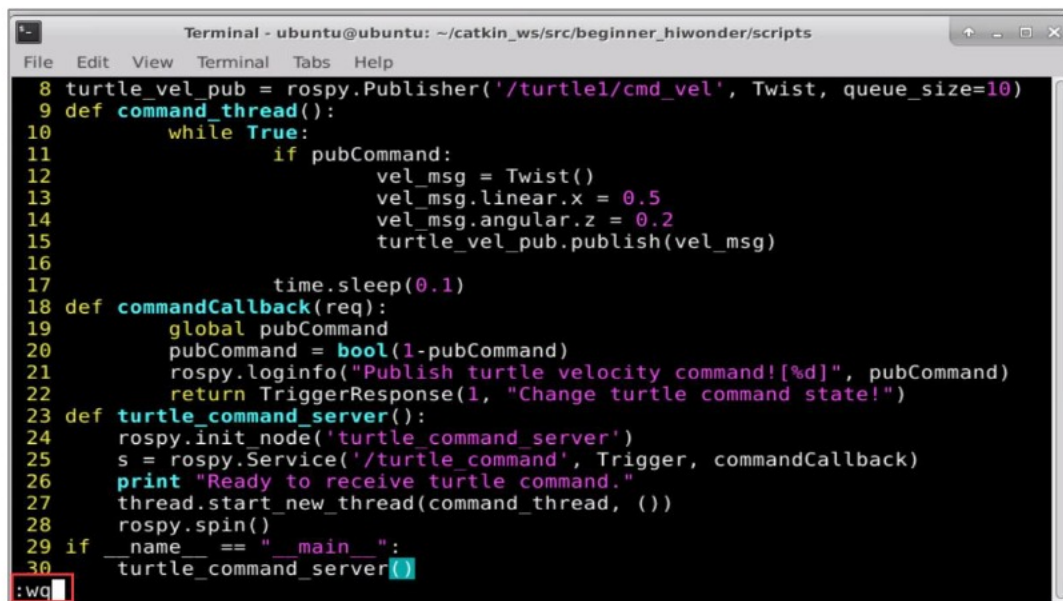
    print "Ready to receive turtle command."

    thread.start_new_thread(command_thread, ())

    rospy.spin()

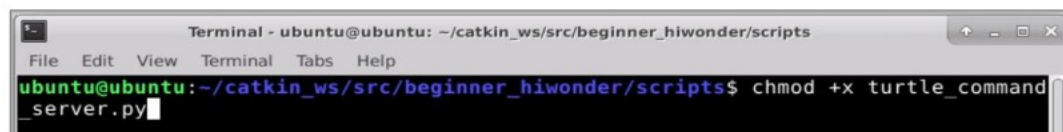
if __name__ == "__main__":

    turtle_command_server()
```



```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder/scripts
File Edit View Terminal Tabs Help
8 turtle_vel_pub = rospy.Publisher('/turtle1/cmd_vel', Twist, queue_size=10)
9 def command_thread():
10     while True:
11         if pubCommand:
12             vel_msg = Twist()
13             vel_msg.linear.x = 0.5
14             vel_msg.angular.z = 0.2
15             turtle_vel_pub.publish(vel_msg)
16
17             time.sleep(0.1)
18 def commandCallback(req):
19     global pubCommand
20     pubCommand = bool(1-pubCommand)
21     rospy.loginfo("Publish turtle velocity command! [%d]", pubCommand)
22     return TriggerResponse(1, "Change turtle command state!")
23 def turtle_command_server():
24     rospy.init_node('turtle_command_server')
25     s = rospy.Service('/turtle_command', Trigger, commandCallback)
26     print "Ready to receive turtle command."
27     thread.start_new_thread(command_thread, ())
28     rospy.spin()
29 if __name__ == "__main__":
30     turtle_command_server()
:wq
```

3) 输入指令“`chmod +x turtle_command_server.py`”，并按下回车，为保存的 `turtle_command_server.py` 赋予可执行权限。



```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder/scripts
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$ chmod +x turtle_command_server.py
```

2.运行服务器节点

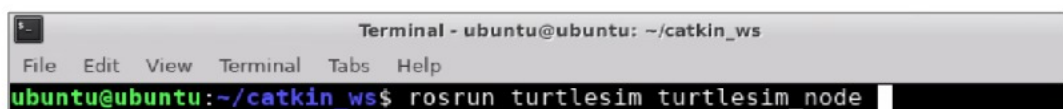
1) 输入指令“`roscore`”，启动节点管理器。

```
ubuntu@ubuntu:~/catkin_ws$ roscore
```

若已开启，则会出现以下提示：

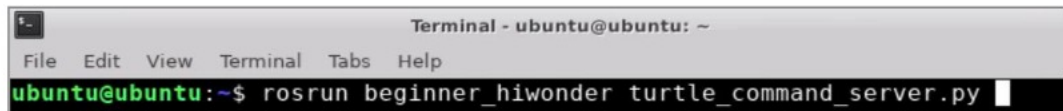
```
RLException: roscore cannot run as another roscore/master is already running.
Please kill other roscore/master processes before relaunching.
The ROS_MASTER_URI is http://ubuntu:11311/
The traceback for the exception was written to the log file
```

2) 输入指令“`roslaunch turtlesim turtlesim_node`”，并按下回车，启动小乌龟仿真器。



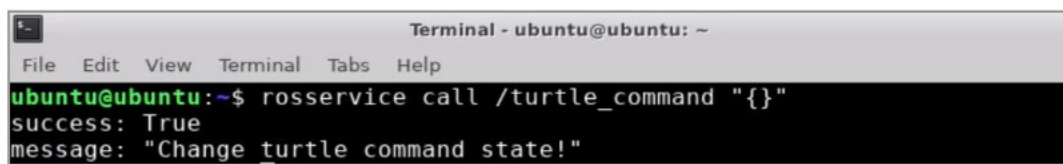
```
Terminal - ubuntu@ubuntu: ~/catkin_ws
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws$ roslaunch turtlesim turtlesim_node
```

3) 打开一个新的命令行终端，输入指令“**roslaunch beginner_hiwonder turtle_command_server.py**”，并按下回车，运行服务器节点。如需停止节点的运行，可按下快捷键“**Ctrl+C**”。



```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ roslaunch beginner_hiwonder turtle_command_server.py
```

4) 再次打开一个新的命令行终端，输入指令“**rosservice call /turtle_command "{}"**”，并按下回车，令小乌龟沿着圆形轨迹移动。



```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ rosservice call /turtle_command "{}"
success: True
message: "Change turtle command state!"
```

